



MHH772 / 90DaysOfDevOps



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LondheShubham153 Added day 30 task

c2e0984 · 7 months ago



This branch is 22 commits ahead of TrainWithShubham/90DaysOfDevOps:master .



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Sync fork

Name

Name

Last commit date



..



README.md

Added day 30 task

7 months ago

README.md



Day 30 – Docker Images & Container Lifecycle

Task

Today's goal is to understand how images and containers actually work.

You will:

- Learn the relationship between images and containers
- Understand image layers and caching
- Master the full container lifecycle

Expected Output

- A markdown file: day-30-images.md

- Screenshots of key commands

Challenge Tasks

Task 1: Docker Images

1. Pull the `nginx`, `ubuntu`, and `alpine` images from Docker Hub
2. List all images on your machine — note the sizes
3. Compare `ubuntu` vs `alpine` — why is one much smaller?
4. Inspect an image — what information can you see?
5. Remove an image you no longer need

Task 2: Image Layers

1. Run `docker image history nginx` — what do you see?
2. Each line is a **layer**. Note how some layers show sizes and some show 0B
3. Write in your notes: What are layers and why does Docker use them?

Task 3: Container Lifecycle

Practice the full lifecycle on one container:

1. **Create** a container (without starting it)
2. **Start** the container
3. **Pause** it and check status
4. **Unpause** it
5. **Stop** it
6. **Restart** it
7. **Kill** it
8. **Remove** it

Check `docker ps -a` after each step — observe the state changes.

Task 4: Working with Running Containers

1. Run an Nginx container in detached mode
2. View its **logs**

3. View **real-time logs** (follow mode)
4. **Exec** into the container and look around the filesystem
5. Run a single command inside the container without entering it
6. **Inspect** the container — find its IP address, port mappings, and mounts

Task 5: Cleanup

1. Stop all running containers in one command
2. Remove all stopped containers in one command
3. Remove unused images
4. Check how much disk space Docker is using

Hints

- Image history: `docker image history`
- Create without starting: `docker create`
- Follow logs: `docker logs -f`
- Inspect: `docker inspect`
- Cleanup: `docker system df` , `docker system prune`

Submission

1. Add your `day-30-images.md` to `2026/day-30/`
2. Commit and push to your fork

Learn in Public

Share what surprised you about image layers or container states on LinkedIn.

#90DaysOfDevOps #DevOpsKaJosh #TrainWithShubham

Happy Learning! **TrainWithShubham**